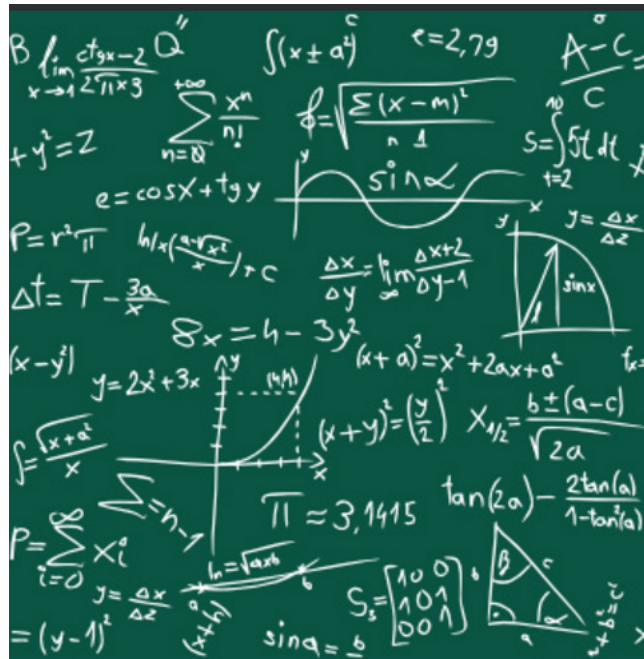


Exercises: properties of sequences



- **Authors:** Fabio Furini and Gianluca Priori.
- **Book:** “Esercitazioni Analisi matematica 1”, Marco Bramanti, Esculapio.

Exercise 1

Consider the sequence

$$a_n = \log\left(1 + (-1)^n \frac{n}{n+1}\right), \quad \text{for } n = 1, 2, 3, \dots$$

Questions:

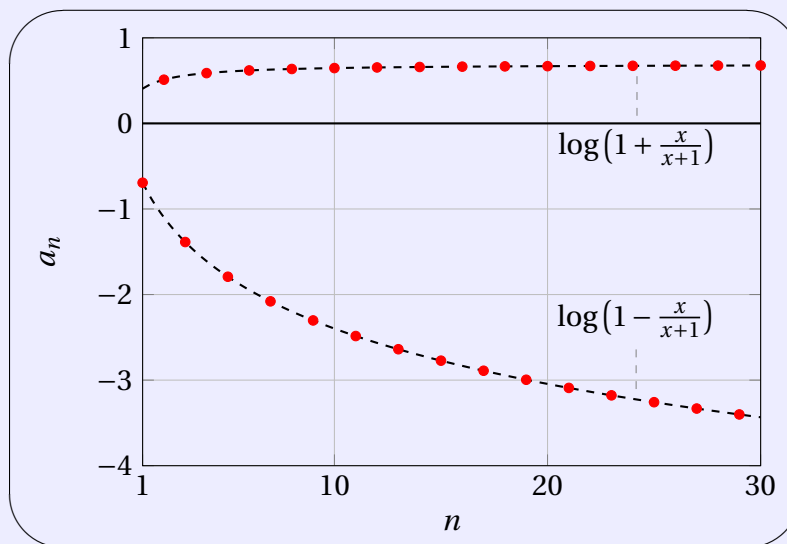
1. Is the sequence bounded above? If so, determine $\sup\{a_n\}$.
2. Does the sequence have a maximum? If so, determine $\max\{a_n\}$.
3. Is the sequence bounded below? If so, determine $\inf\{a_n\}$.
4. Does the sequence have a minimum? If so, determine $\min\{a_n\}$.

For $n = 1$ we have

$$\log\left(1 - 1 \frac{1}{2}\right) = \log \frac{1}{2} \approx -0.69314.$$

For $n = 2$ we have

$$\log\left(1 + 1 \frac{2}{3}\right) = \log \frac{5}{3} \approx 0.510826.$$



The presence of the alternating sign $(-1)^n$ suggests studying the behavior of the sequence by distinguishing what happens for even n and odd n .

- If n is even:

$$a_n = \log\left(1 + \frac{n}{n+1}\right), \quad \text{for } n = 2, 4, 6 \dots$$

hence the argument of the logarithm is

$$1 + \frac{n}{n+1} = 1 + \frac{n+1-1}{n+1} = 2 - \frac{1}{n+1}$$

which varies in

$$\left[\frac{5}{3}, 2\right) \text{ and the logarithm in } \left[\log \frac{5}{3}, \log 2\right),$$

i.e., it is positive and bounded above. Hence

$$\sup \{a_n : n \text{ is even}\} = \log 2 \approx 0.69314$$

but $\log 2$ is not the maximum of these values.

- If n is odd:

$$a_n = \log\left(1 - \frac{n}{n+1}\right), \quad \text{for } n = 1, 3, 5 \dots$$

hence the argument of the logarithm is

$$1 - \frac{n}{n+1} = \frac{1}{n+1}$$

which varies in

$$\left(0, \frac{1}{2}\right] \text{ and the logarithm in } \left(-\infty, \log \frac{1}{2}\right],$$

i.e., it is negative and, as n increases, it is unbounded below.

The sequence as a whole is therefore bounded above but not bounded below; it has no maximum (although its sup is finite) and no minimum (because it is unbounded below).

Exercise 2

Consider the sequence

$$a_n = e^{-\frac{1}{n}} \sin n, \quad \text{for } n = 1, 2, 3, \dots$$

Questions:

1. Is the sequence bounded above?
2. Is the sequence bounded below?
3. Is it eventually positive?
4. Does it never vanish?
5. Does it have a limit (finite or infinite)?

For $n = 1$ we have

$$e^{-1} \sin 1 \approx 0.3679 \cdot 0.8415 \approx 0.309560.$$

For $n = 2$ we have

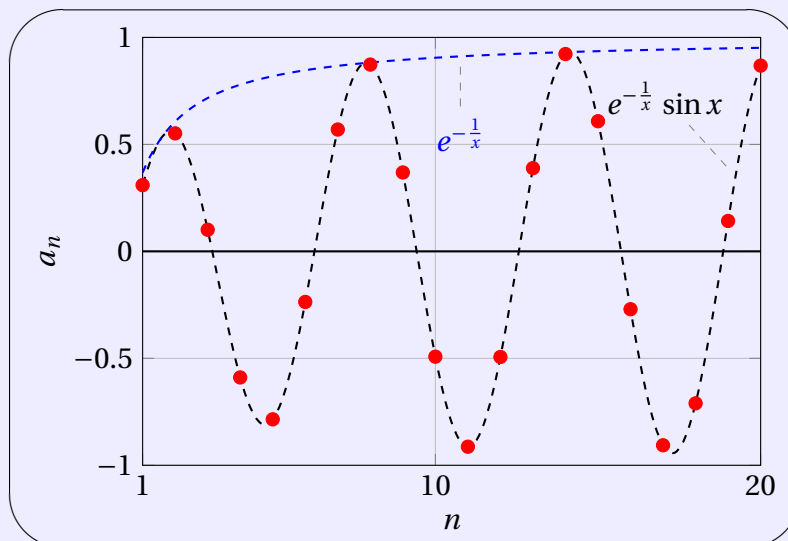
$$e^{-\frac{1}{2}} \sin 2 \approx 0.6065 \cdot 0.9093 \approx 0.551517.$$

For $n = 10$ we have

$$e^{-\frac{1}{10}} \sin 10 \approx 0.9048 \cdot -0.5440 \approx -0.492251.$$

For $n = 20$ we have

$$e^{-\frac{1}{20}} \sin 20 \approx 0.9512 \cdot 0.9129 \approx 0.8684.$$



The sequence is the product of the sequence

$$n \mapsto e^{-\frac{1}{n}} \text{ with } \lim_{n \rightarrow +\infty} e^{-\frac{1}{n}} = 1,$$

which is bounded, always nonzero, convergent, and of the sequence

$$n \mapsto \sin n,$$

which is bounded but irregular, and never zero (since n starts from 1, by the irrationality of π , the angle n is never an integer multiple of π).

Hence, the sequence given by the product of these two is bounded, never zero, and irregular.

Since

$$e^{-\frac{1}{n}} \rightarrow 1$$

and the sign of $\sin n$ is not eventually constant, the sequence given by the product of the two is not eventually positive.

Exercise 3

Consider the sequence

$$a_n = e^n \sin n, \quad \text{for } n = 1, 2, 3, \dots$$

Questions:

1. Is the sequence bounded above?
2. Is the sequence bounded below?
3. Is it eventually positive?
4. Does it never vanish?
5. Does it have a limit (finite or infinite)?

For $n = 1$ we have

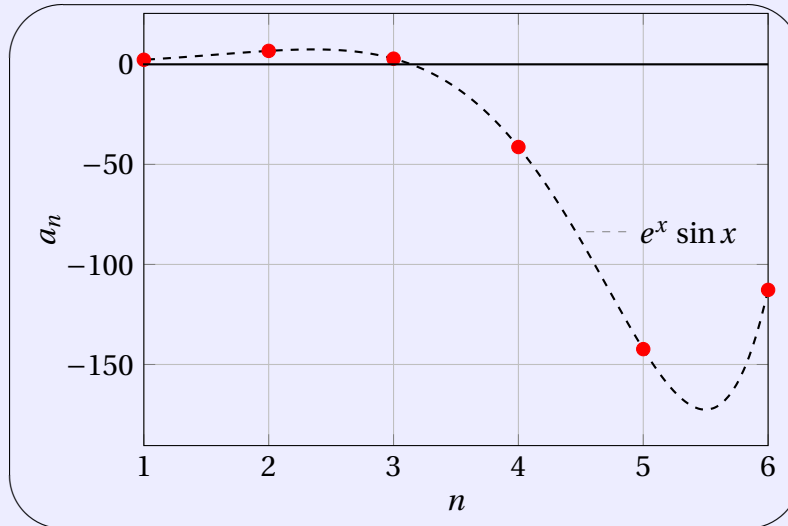
$$e^1 \sin 1 \approx 2.7183 \cdot 0.8415 \approx 2.2874.$$

For $n = 2$ we have

$$e^2 \sin 2 \approx 7.3891 \cdot 0.9093 \approx 6.7188.$$

For $n = 5$ we have

$$e^5 \sin 5 \approx 148.4132 \cdot -0.9589 \approx -142.3170.$$



The sequence is the product of the sequence

$$n \mapsto e^n \text{ with } \lim_{n \rightarrow +\infty} e^n = +\infty,$$

which is unbounded above, bounded below, always nonzero, divergent, and of the sequence

$$n \mapsto \sin n,$$

which is bounded but irregular, and never zero (since n starts from 1, by the irrationality of π , the angle n is never an integer multiple of π).

Hence, the sequence given by the product of these two is unbounded above, unbounded below, never zero, and irregular.

Since

$$e^n \rightarrow +\infty$$

and the sign of $\sin n$ is not eventually constant, the sequence given by the product of the two is not eventually positive.

Exercise 4

Consider the sequence

$$a_n = \frac{n^{(-1)^n}}{n+1}, \quad \text{for } n = 1, 2, 3, \dots$$

Questions:

1. Is the sequence bounded above? If so, determine $\sup \{a_n\}$.
2. Does the sequence have a maximum? If so, determine $\max \{a_n\}$.
3. Is the sequence bounded below? If so, determine $\inf \{a_n\}$.
4. Does the sequence have a minimum? If so, determine $\min \{a_n\}$.

For $n = 1$ we have

$$1^{(-1)^1} \frac{1}{2} = 1 \frac{1}{2}$$

For $n = 2$ we have

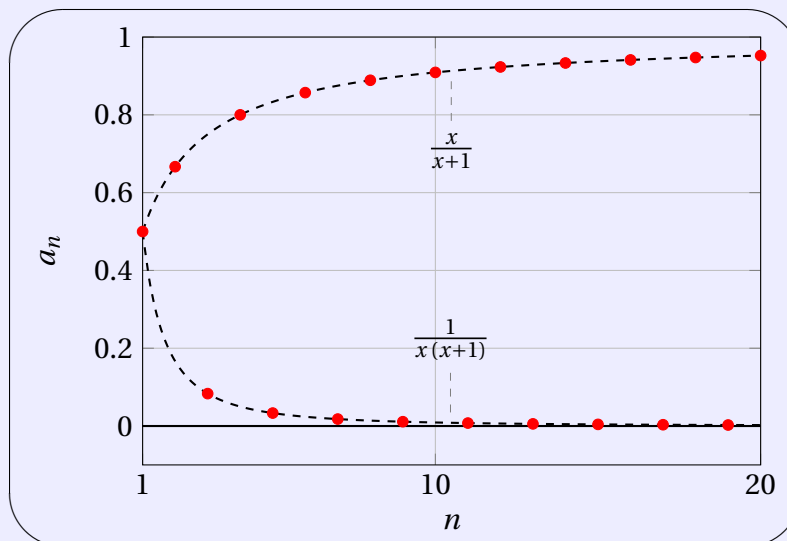
$$2^{(-1)^2} \frac{1}{3} = 2 \frac{1}{3}.$$

For $n = 3$ we have

$$3^{(-1)^3} \frac{1}{4} = \frac{1}{3} \frac{1}{4}.$$

For $n = 4$ we have

$$4^{(-1)^4} \frac{1}{5} = 4 \frac{1}{5}.$$



The presence of the alternating sign $(-1)^n$ suggests studying the behavior of the sequence by distinguishing what happens for even n and odd n .

- If n is even:

$$a_n = \frac{n}{n+1}, \quad \text{for } n = 2, 4, 6, \dots$$

Adding $+1$ and -1 to the numerator, we obtain

$$a_n = \frac{n}{n+1} = \frac{n+1-1}{n+1} = 1 - \frac{1}{n+1}$$

which varies in

$$\left[\frac{2}{3}, 1 \right),$$

i.e., it is positive and bounded above. Hence

$$\sup \{a_n : n \text{ is even} \} = 1$$

but 1 is not the maximum of these values.

- If n is odd:

$$a_n = \frac{\frac{1}{n}}{n+1} = \frac{1}{n} \cdot \frac{1}{n+1} = \frac{1}{n(n+1)}, \quad \text{for } n = 1, 3, 5, \dots$$

which varies in

$$\left(0, \frac{1}{2} \right],$$

i.e., it is positive and bounded below. Hence

$$\inf \{a_n : n \text{ is odd} \} = 0$$

but 0 is not the minimum of these values.

The sequence as a whole is bounded; it has no maximum (although its sup is finite) and no minimum (although its inf is finite).